QI LING

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3rd-year Ph.D. student in Computer Science, Purdue University

RESEARCH INTERESTS

I want to make computer systems more efficient, secure, and reliable. Currently, my work focuses on fine-grained and low-cost microarchitectural isolation techniques, enhancing performance and security for serverless computing. Still, my interest spans many other fields in computer architecture and operating systems.

EDUCATION

- **Purdue University** **Aug. 2024 – Present**
  Doctoral Degree in CS **Overall GPA: 4.00/4**
  Advisors: Prof. Kazem Taram and Prof. Pedro Fonseca
- **University of Michigan** **Sept. 2022 – May 2024**
  Bachelor's Degree in CS **Overall GPA: 3.92/4**
  Advisors: Prof. Baris Kasikci and Prof. Shuwen Deng
- **Shanghai Jiao Tong University** **Sept. 2020 – Aug. 2024**
  Bachelor's Degree in ECE **Overall GPA: 3.74/4**
  Advisor: Prof. Yibo Pi

AWARDS AND HONORS

- **USENIX Security Distinguished Paper Award** **2026**
- **ACM MICRO Student Research Competition 2nd Place Award** **2023**
- **Dean's Honor List** **2022, 2023**
- **Jackson and Murial Lum Scholarship** **2022, 2023**
- **SJTU Undergraduate Excellence Scholarship** **2021**

SELECTED PUBLICATIONS

- **A Harmony of Ice and Fire: Mitigating Cold Last-Level Cache Effects in Serverless Computing on Commodity Hardware**
  Qi Ling, Ajay R. Rawat, Pedro Fonseca, Kazem Taram
  59th IEEE/ACM International Symposium on Microarchitecture (**MICRO'26**)
- **An Integer Overflow Endgame: Compiler and Architecture Support for Default-On Integer Overflow Mitigation**
  Zheng Zhang, **Qi Ling**, Kian Kasad, Brendan Ryan Sweeney, Deepak Gupta, Tal Garfinkel, Kazem Taram
  35th USENIX Security Symposium (**USENIX Security'26**)
  Distinguished Paper Award
- **GadgetMeter: Quantitatively and Accurately Gauging the Exploitability of Speculative Gadgets**
  Qi Ling, Yujun Liang, Yi Ren, Baris Kasikci, Shuwen Deng
  Network and Distributed System Security Symposium (**NDSS'25**)
  2nd Place at MICRO Student Research Competition

TECHNICAL SKILLS

- **CPU design**: Verilog, Gem5
  - Implemented a P6 OoO processor with speculative execution and prefetchers.
- **CPU microarchitectural analysis**
  - Identified program bottlenecks in CPU microarchitecture.
  - Characterized Spectre vulnerabilities and microarchitectural side channels.
- **LLVM compiler development**
  - Added frontend and backend support for new overflow instructions.
  - Developed middle-end passes to accurately identify and patch Spectre vulnerabilities.
- **Linux kernel development**
  - Implemented page coloring (1k LoC) within Linux memory management subsystem.
  - Implemented an ultra-fast NAT router with an eBPF module (GitHub).
- **Others**: Honggfuzz, Syzkaller, C/C++, Python, assembly, Kubernetes

RESEARCH EXPERIENCE

- **SecArch & RSSys, Purdue University** **Aug. 2024 - Present**
  Advisors: Kazem Taram and Pedro Fonseca
  Microarchitectural isolation for performance and security.
  - Identified a new kind of cache inefficiency in serverless computing.
  - Discovered a more fine-grained and low-cost cache partitioning technique, without introducing new hardware and applicable to all recent Intel CPUs.
  - Proposed three guiding principles to achieve efficient cache management across tens to hundreds of serverless functions.
  - Improved the performance of latency-critical functions by up to 65% and 20% on average.
- **EFESLAB, University of Michigan** **Dec. 2022 - Dec. 2023**
  Advisors: Baris Kasikci and Shuwen Deng
  Detecting and evaluating Spectre-PHT gadgets in programs.
  - Identified the **Timing Condition** of Spectre-PHT gadget.
  - Presented a systematic study and exploration of windowing power.
  - Proposed and implemented a new approach, which describes the timing condition with a DAG graph, models the windowing power with graph algorithms, and quantifies the gadget's exploitability with runtime measurement.
  - Validated our approach and evaluated it against SOTA scanners on 2 micro-benchmarks, 6 userspace programs, and the Linux kernel. Identify 503 gadgets reported by SOTA scanners as unexploitable.

ACADEMIC SERVICE & TEACHING

- **USENIX Security Symposium Program Committee** **2027**
- **Computer Architecture Teaching Assistant** **Fall 2026, Fall 2024**
- **Computer Algorithm Teaching Assistant** **Spring 2026**
- **Computer Security Teaching Assistant** **Fall 2025, Spring 2025**